

Sneha Agarwal

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EDUCATION

University of California, Los Angeles - UCLA	Los Angeles, CA
M.S. in Computer Science	Sep 2027 - June 2028
B.S. in Computer Science, Minor in Data Science Engineering GPA: 3.92 Dean's Honor List	Sep 2023 - June 2027
• Relevant Coursework: Deep Learning and Neural Networks, Natural Language Processing, Machine Learning, Data Structures & Algorithms, Algorithms and Complexity, Operating Systems, Computer Networks, Software Construction, Computer Organization, Databases, Linear Algebra	

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, TensorFlow), C, C++, Java, React, HTML/CSS, JavaScript, TypeScript, SQL
Tools & Technologies: AWS, Databricks, Dask, Flask, Git, Docker, MongoDB, PostgreSQL, HuggingFace, LangGraph, Axolotl, VSCode, Claude Code

WORK EXPERIENCE

AI/ML Engineer Intern Adobe San Jose, CA	June 2026 - Sep 2026
• Built, fine-tuned, and deployed a prompt-tunable fast-path SLM router for Acrobat AI Assistant agentic harness using Axolotl, AWS SageMaker, and vLLM, routing simple queries directly to specialized agents; reduced routing latency 80%, improved recall 21%, and reduced misroutes 58%. • Developed a reusable model-distillation and SLM training framework with automated synthetic-data generation and evaluation using NVIDIA Data Designer, enabling scalable training for future SLM use cases; trained across diverse agent rosters and output schemas to enable routing-policy changes without data regeneration or model retraining.	
Software Engineer Intern Apple Cambridge, MA	June 2025 - Sep 2025
• Researched and integrated diverse text data sources into an end-to-end NLP and anomaly-detection pipeline to identify notable weather conditions. • Processed large-scale datasets with Dask and AWS infrastructure for scalable data processing and model training. Experimented with BERT, Sentence Transformers, NER, and LLM-based extraction to extract insights from free-form text data. • Trained an XGBoost severe-weather classifier achieving 90%+ accuracy; performed feature engineering and SHAP analysis to identify key weather features and thresholds.	
Software Engineer Intern Juniper Networks Sunnyvale, CA	June 2024 - Sep 2024
• Automated code-review training-data creation using Python scripting; generated synthetic Q&A data from product documentation with LangChain chunking and API integration. Fine-tuned Llama 3 with LoRA, achieving 95% accuracy for code reviews and Q&A. • Designed and implemented a prompt-chaining pipeline to customize the model for coding guidelines and built an LLM Routing Engine to direct user queries to various APIs, including LLM chat, NLQ for SQL queries, and RAG vector database based on query intent.	

PROJECTS & RESEARCH

NLP Researcher @ UCLA MARS Lab	Mar 2026 - June 2026
• Analyzed SLM agent trajectories across SWE-ZERO, Endless Terminals, and Terminal-Bench to identify failure modes in coding and terminal tasks. • Built heuristic and LLM-as-judge evaluation pipelines to develop a failure-mode taxonomy and labeling rubric, informing dataset design and training recommendations for more reliable agentic SLMs.	
AskBruin Amazon Web Services (IAM, S3, Cognito, Lambda, Bedrock), React, APIs	July 2025 - Mar 2026
• Built a multi-agent AI assistant to centralize UCLA data and answer queries across events, course planning, and campus resources. • Developed automated ingestion, embedding, and RAG pipelines using AWS Lambda, S3, Bedrock, and vector databases to process unstructured event data and persist conversational sessions.	
LA Fire Relief Insurance Household Item Tracker + Policy Analyzer Flask, React, Azure, APIs	Jan 2025 - Mar 2025
• Developed a Flask backend and React frontend that parsed Amazon orders and credit-card statements to automate insurance-claim inventory creation. • Implemented a vector database of insurance documents for RAG using the OpenAI API, storing embeddings in Azure Assistant Infrastructure to enhance claim-policy matching; integrated the Perplexity API for search and summarization of insurance plans.	
EMG Keystroke Prediction PyTorch, Numpy, Tensorboard	Feb 2025 - Mar 2025
• Built and evaluated CNN, LSTM, GRU, and Transformer architectures on the emg2qwerty dataset to map sEMG signals to keystrokes using CTC loss; achieved over a 20% reduction in CER compared to the baseline model. • Applied and benchmarked temporal data augmentation techniques (e.g., random resampling, masking), and conducted hyperparameter tuning.	

LEADERSHIP EXPERIENCE & CLUB INVOLVEMENT

Director of External Events, AI/ML Engineer Bruin AI UCLA	Oct 2024 - Present
• Led Bruin AI external programming for 400+ attendee events, managing teams across logistics, speaker panels, workshops, and mentorship. • Developed AI-powered vendor intelligence tools including spend dashboards, purchase search, document comparison, and consolidation insights.	
Data Scientist and Project Manager Data Science Union UCLA	Apr 2024 - Present
• Led a team of 5 in building a spatiotemporal graph neural network (STGNN) pipeline for real-time traffic prediction and route optimization. • Integrated OpenStreetMap, TomTom, and PeMS traffic data; developed a GRU + GCN + Transformer architecture to model spatial and temporal dependencies and predict optimal routes.	
Internal Board Treasurer Society of Women Engineers UCLA	Oct 2023 - June 2025
ACM Hack Officer Association for Computing Machinery UCLA	Oct 2023 - Present